Load data

In [1]:
```
import pandas as  pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns


df = pd.read_csv('ecommerce_sales_data.csv')
```

EDA

In [2]:
```
df.isnull().sum()
```

Out[2]:
```
TransactionID           0
CustomerID              0
ProductID               0
ProductCategory         0
Quantity                0
Price                   0
Discount                0
TransactionDate         0
PaymentMethod           0
CustomerLocation        0
CustomerAge             0
CustomerGender          0
CustomerIncomeGroup     0
CustomerLoyaltyScore    0
dtype: int64
```

In [3]:
```
df.info()
df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 14 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   TransactionID         100000 non-null  int64
 1   CustomerID            100000 non-null  int64
 2   ProductID             100000 non-null  int64
 3   ProductCategory       100000 non-null  str
 4   Quantity              100000 non-null  int64
 5   Price                 100000 non-null  float64
 6   Discount              100000 non-null  float64
 7   TransactionDate       100000 non-null  str
 8   PaymentMethod         100000 non-null  str
 9   CustomerLocation      100000 non-null  str
 10  CustomerAge           100000 non-null  int64
 11  CustomerGender        100000 non-null  str
 12  CustomerIncomeGroup   100000 non-null  str
 13  CustomerLoyaltyScore  100000 non-null  float64
dtypes: float64(3), int64(5), str(6)
memory usage: 16.3 MB
```

Out[3]:

	TransactionID	CustomerID	ProductID	Quantity	Price	
count	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100
mean	50000.500000	5504.000280	550.554140	2.499750	501.469993	
std	28867.657797	2600.472561	260.256267	1.118353	287.536905	
min	1.000000	1000.000000	100.000000	1.000000	5.010000	
25%	25000.750000	3251.000000	325.000000	1.000000	251.537500	
50%	50000.500000	5511.000000	552.000000	3.000000	501.580000	
75%	75000.250000	7751.000000	777.000000	3.000000	750.397500	
max	100000.000000	9999.000000	999.000000	4.000000	1000.000000	

In [4]:
```
df.drop(columns=['TransactionID', 'CustomerID','ProductID'], inplace=True)
```

In [5]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 11 columns):
 #   Column                 Non-Null Count   Dtype
---  ------                 --------------   -----
 0   ProductCategory        100000 non-null  str
 1   Quantity               100000 non-null  int64
 2   Price                  100000 non-null  float64
 3   Discount               100000 non-null  float64
 4   TransactionDate        100000 non-null  str
 5   PaymentMethod          100000 non-null  str
 6   CustomerLocation       100000 non-null  str
 7   CustomerAge            100000 non-null  int64
 8   CustomerGender         100000 non-null  str
 9   CustomerIncomeGroup    100000 non-null  str
 10  CustomerLoyaltyScore   100000 non-null  float64
dtypes: float64(3), int64(2), str(6)
memory usage: 14.0 MB
```

In [6]:
```
df.duplicated().sum()
```

Out[6]:
```
np.int64(0)
```

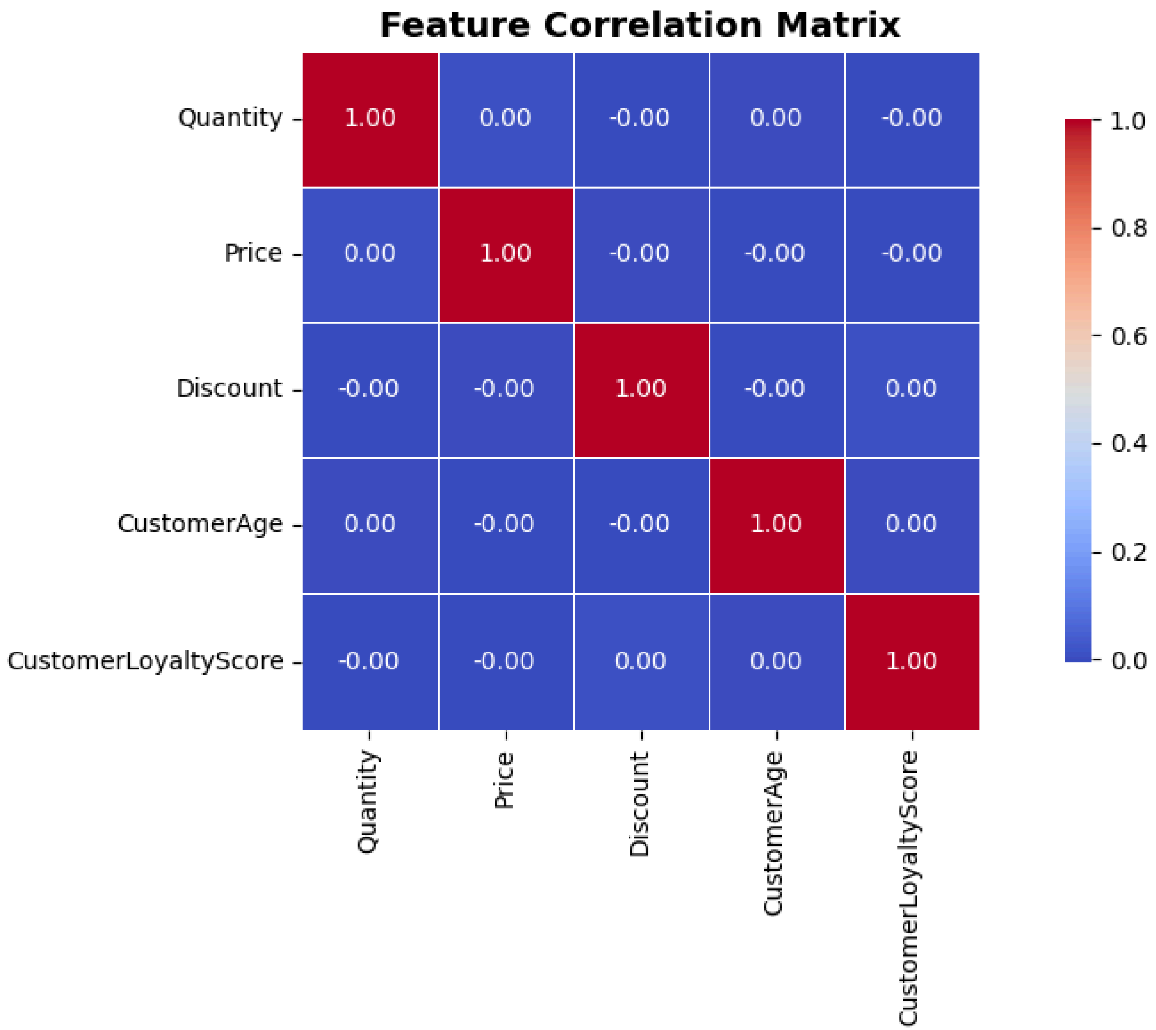
In [7]:
```
df.dtypes
```

Out[7]:
```
ProductCategory             str
Quantity                  int64
Price                   float64
Discount                float64
TransactionDate             str
PaymentMethod               str
CustomerLocation            str
CustomerAge               int64
CustomerGender              str
CustomerIncomeGroup         str
CustomerLoyaltyScore    float64
dtype: object
```

In [8]:
```
df['TransactionDate'] = pd.to_datetime(df['TransactionDate'])
```

Graphs and insights

In [9]:
```
#heatmap
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, fmt='.2f', cmap='coolwarm',
            linewidths=0.5, square=True, cbar_kws={"shrink": 0.8})
plt.title('Feature Correlation Matrix', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```



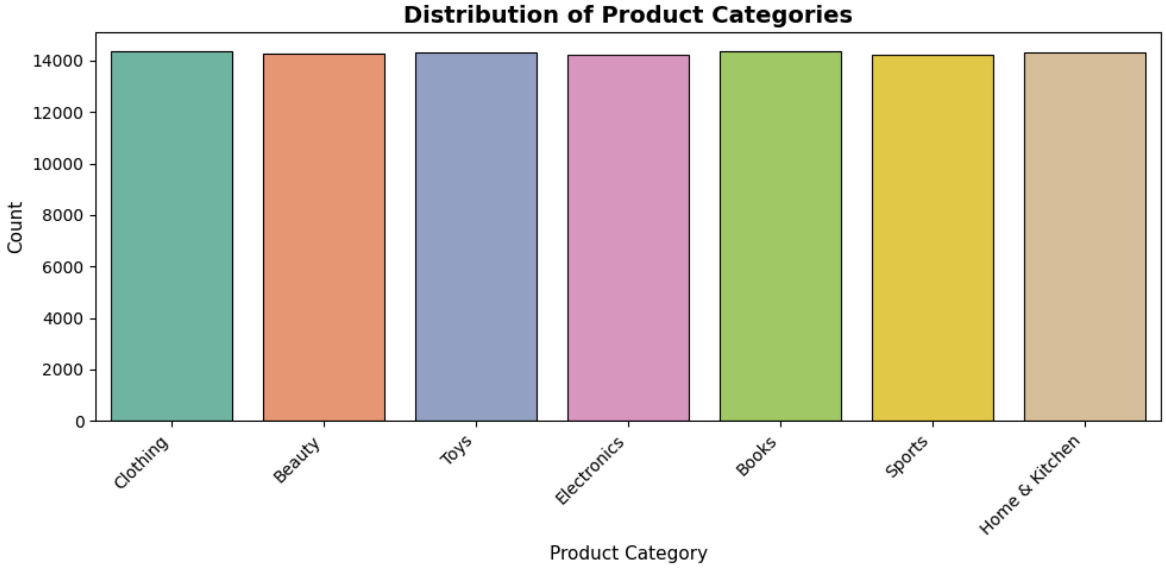
> The correlation analysis indicates that most numerical variables in the dataset have very weak relationships with each other, suggesting independent behavior among customer and transaction features.

In [10]:
```
# 2. Product Category Countplot
plt.figure(figsize=(10,5))
sns.countplot(x='ProductCategory', data=df, palette='Set2', edgecolor='black', l
plt.title('Distribution of Product Categories', fontsize=14, fontweight='bold')
plt.xlabel('Product Category', fontsize=11)
plt.ylabel('Count', fontsize=11)
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
```

```
C:\Users\wardo\AppData\Local\Temp\ipykernel_22360\770576662.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.countplot(x='ProductCategory', data=df, palette='Set2', edgecolor='black',
linewidth=0.8)
```



> Product Category Distribution Insights

> The distribution of product categories is almost uniform across all categories.

> Clothing, Books, Toys, Electronics, Beauty, Sports, and Home & Kitchen have nearly equal transaction counts.

> No single category dominates the dataset, which indicates balanced customer purchasing behavior across categories.

> This balanced distribution is useful for unbiased analysis and machine learning modeling.

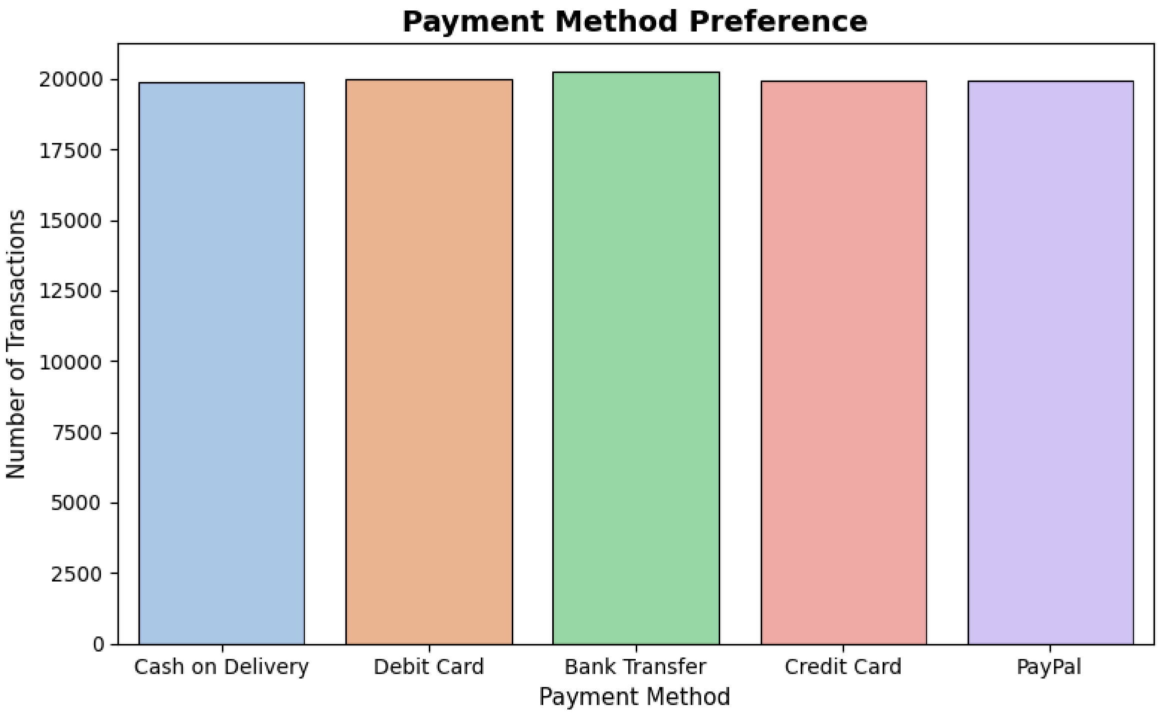
In [11]:

```
# 3. Payment Method Countplot
plt.figure(figsize=(8,5))
sns.countplot(x='PaymentMethod', data=df, palette='pastel', edgecolor='black', l
plt.title('Payment Method Preference', fontsize=14, fontweight='bold')
plt.xlabel('Payment Method', fontsize=11)
plt.ylabel('Number of Transactions', fontsize=11)
plt.tight_layout()
```

```
C:\Users\wardo\AppData\Local\Temp\ipykernel_22360\1398334644.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.countplot(x='PaymentMethod', data=df, palette='pastel', edgecolor='black',
linewidth=0.8)
```



> Payment Method Preference Insights

> Customers are using all payment methods almost equally.

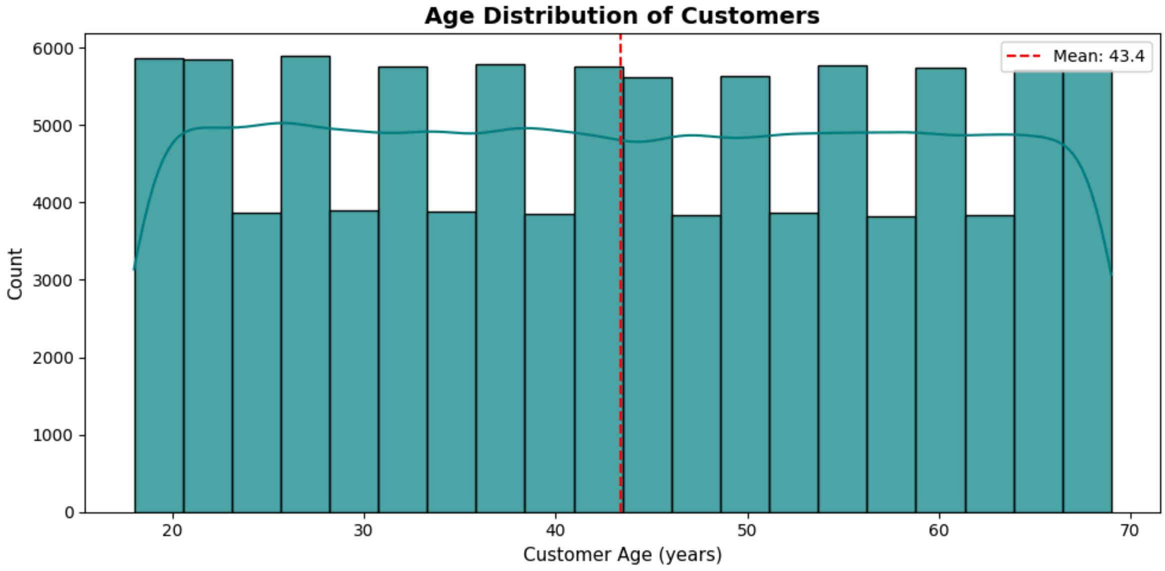
> Bank Transfer appears to have slightly higher usage compared to other payment methods.

> Cash on Delivery, Credit Card, Debit Card, and PayPal transactions are also consistently high.

> The dataset shows diversified payment behavior with no heavy dependency on a single payment method.

In [12]:

```
# 4. Customer Age Histogram
plt.figure(figsize=(10,5))
sns.histplot(df['CustomerAge'], bins=20, kde=True, color='teal', alpha=0.7, edge
plt.title('Age Distribution of Customers', fontsize=14, fontweight='bold')
plt.xlabel('Customer Age (years)', fontsize=11)
plt.ylabel('Count', fontsize=11)
plt.axvline(df['CustomerAge'].mean(), color='red', linestyle='--', label=f'Mean:
plt.legend()
plt.tight_layout()
```



> Customer Age Distribution Insights

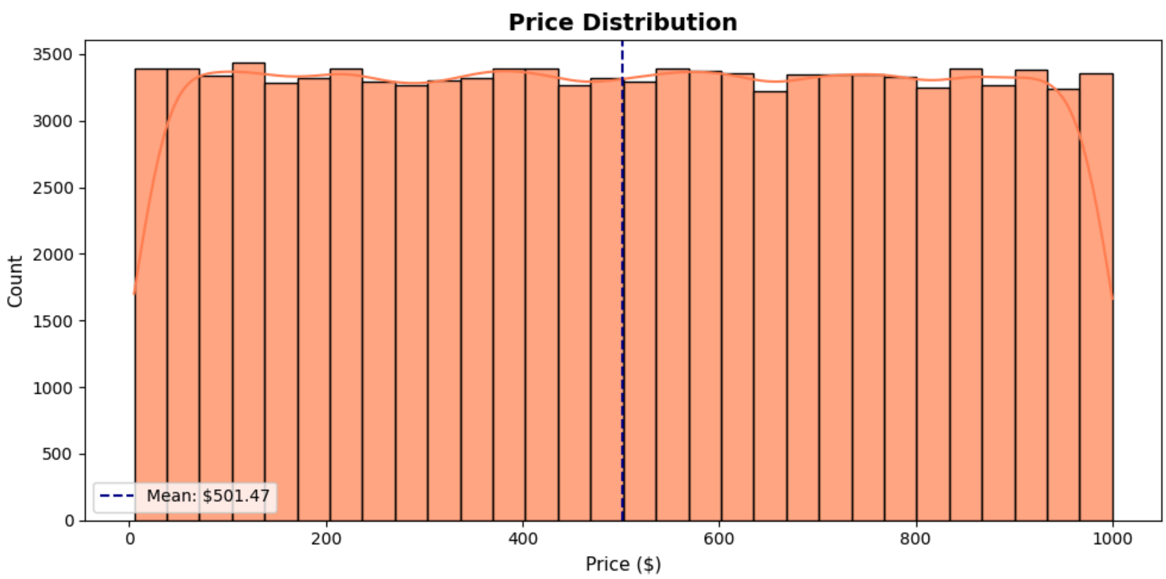
> Customer ages are distributed fairly evenly between 18 and 70 years.

> The average customer age is approximately 43 years.

> There is no major age concentration or skewness observed in the dataset.

> The business appears to target customers from multiple age groups rather than focusing on a specific demographic.

In [13]:
```
# 5. Price Histogram
plt.figure(figsize=(10,5))
sns.histplot(df['Price'], bins=30, kde=True, color='coral', alpha=0.7, edgecolor
plt.title('Price Distribution', fontsize=14, fontweight='bold')
plt.xlabel('Price ($)', fontsize=11)
plt.ylabel('Count', fontsize=11)
plt.axvline(df['Price'].mean(), color='darkblue', linestyle='--', label=f'Mean:
plt.legend()
plt.tight_layout()
```



> Price Distribution Insights

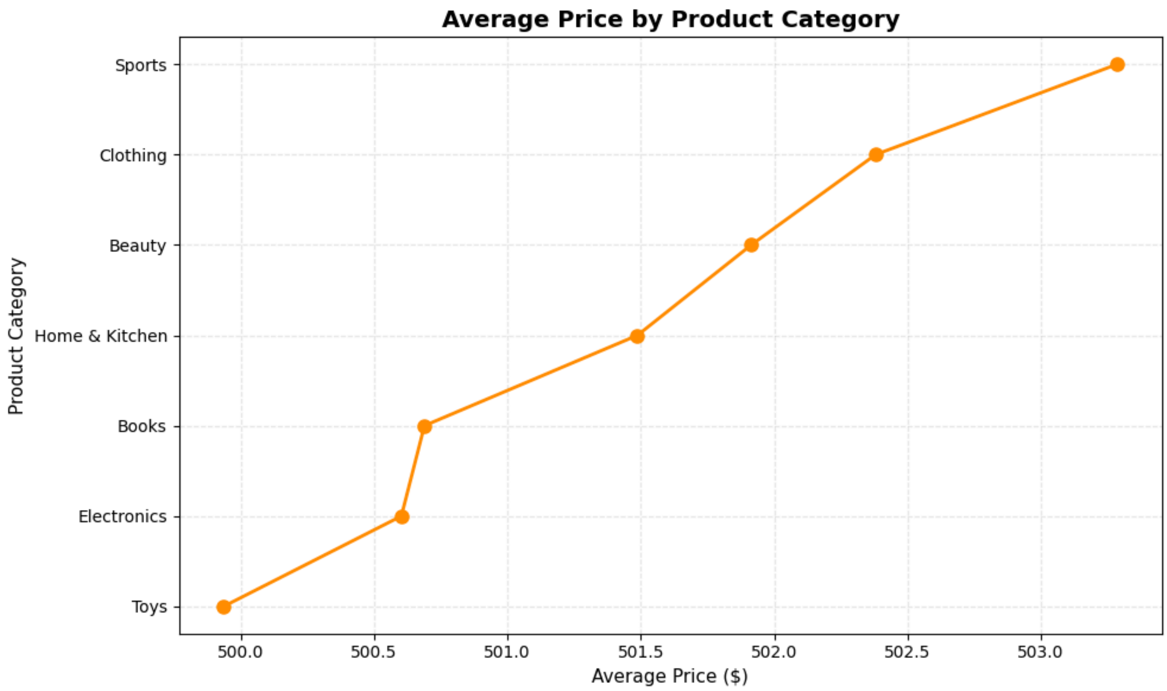
> Product prices are spread almost uniformly between low-price and high-price ranges.

> The average product price is around $501.47.

> No strong skewness or abnormal clustering is observed in product pricing.

> The dataset includes products from multiple pricing segments, making it suitable for comprehensive sales and customer behavior analysis.

In [14]:
```
# Product Category vs Average Price
plt.figure(figsize=(10,6))
avg_price = df.groupby('ProductCategory')['Price'].mean().sort_values()
plt.plot(avg_price.values, avg_price.index, 'o-', color='darkorange', linewidth=
plt.title('Average Price by Product Category', fontsize=14, fontweight='bold')
plt.xlabel('Average Price ($)', fontsize=11)
plt.ylabel('Product Category', fontsize=11)
plt.grid(True, alpha=0.3, linestyle='--')
plt.tight_layout()
```



> Average Price by Product Category Insights

> Sports category has the highest average product price among all categories.

> Toys category has the lowest average product price.

> Most categories have average prices close to $500, indicating a balanced pricing structure across the platform.

> The small variation in average prices suggests that products are distributed evenly across categories without major pricing bias.

In [15]:
```
sns.scatterplot(x='CustomerAge',
        y='CustomerLoyaltyScore',
        data=df
    )

plt.title("Customer Age vs Loyalty Score")
plt.show()
```

> Customer Age vs Loyalty Score Insights

> There is no strong relationship between customer age and loyalty score.

> Customers of all age groups show both high and low loyalty scores.

> Loyalty behavior appears to be independent of customer age in this dataset.

> This suggests that factors other than age may influence customer loyalty more significantly.

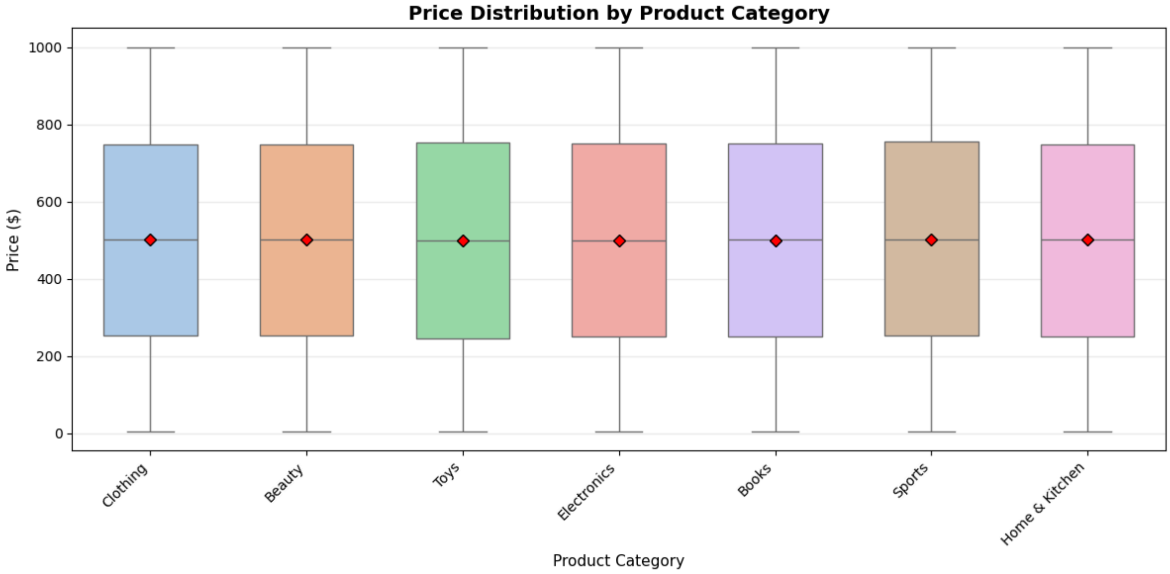
In [16]:
```
plt.figure(figsize=(12,6))
sns.boxplot(x='ProductCategory', y='Price', data=df, palette='pastel', width=0.6
    showmeans=True, meanprops={"marker":"D", "markerfacecolor":"red", "markeredg
plt.title('Price Distribution by Product Category', fontsize=14, fontweight='bol
plt.xlabel('Product Category', fontsize=11)
```

```
plt.ylabel('Price ($)', fontsize=11)
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', alpha=0.2)
plt.tight_layout()
```

```
C:\Users\wardo\AppData\Local\Temp\ipykernel_22360\2796103620.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.boxplot(x='ProductCategory', y='Price', data=df, palette='pastel', width=0.
6,
```



> Price Distribution by Product Category Boxplot Insights

> All product categories show a very similar price distribution pattern.

> Median prices across categories are close to $500.

> The interquartile ranges are also similar, indicating consistent pricing spread across categories.

> No category shows extreme outliers or abnormal price behavior.

> The dataset appears well-balanced, which is beneficial for comparative analysis and machine learning tasks.

In [17]:
```
df.head()
```

Out[17]:

	ProductCategory	Quantity	Price	Discount	TransactionDate	PaymentMethod	Cus
0	Clothing	1	801.66	0.13	2023-12-22 10:35:26	Cash on Delivery	
1	Beauty	1	225.03	0.06	2024-05-05 10:40:33	Debit Card	
2	Toys	2	544.99	0.24	2023-12-04 05:22:59	Debit Card	
3	Beauty	2	157.02	0.01	2024-01-02 11:11:00	Bank Transfer	
4	Electronics	2	155.51	0.14	2024-06-18 01:01:29	Debit Card	

Overall Insights

> Customer purchasing activity is evenly distributed across all product categories, indicating balanced market demand without heavy dependency on a single category.

> The platform serves customers from diverse age groups, showing broad customer engagement across different demographics.

> Product pricing remains relatively stable across categories, suggesting a consistent pricing strategy throughout the business.

> Multiple payment methods are actively used by customers, which reflects flexibility and convenience in the purchasing process.

> Customer loyalty does not appear to be strongly influenced by age, indicating that loyalty behavior may depend more on user experience, product quality, or service factors.

> No strong correlation was observed among major numerical variables such as price, quantity, discount, and loyalty score, suggesting that customer purchasing behavior is influenced by multiple independent factors.

> The dataset is highly balanced with minimal bias in categories, pricing, and customer distribution, making it suitable for reliable exploratory analysis and machine learning implementation.

> Overall, the business demonstrates stable transactional behavior, diversified customer interaction, and consistent product engagement across the platform.